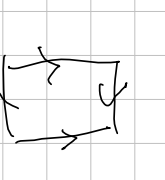


$(\mathbb{Z} \times \mathbb{Z})$
 $aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$

 $S^1 \vee S^1$



 $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$
 $T = \mathbb{Z} \times \mathbb{Z}$
 $T \simeq \mathbb{Z} \times \mathbb{Z}$
 $\mathbb{Z} \times \mathbb{Z}$

[illegible][illegible]

$$Z = \frac{1}{z} + 4$$

$$K_1 = \frac{1}{z}$$

$$K_2 = \frac{1}{z^2}$$

$$2\pi i \cdot \frac{1}{2} \cdot \frac{1}{2} = \pi i$$